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Fields of Concentration:
Microeconomic Theory
Industrial Organization

Qualifying Examinations Completed:
2023: Microeconomic Theory (*with distinction)*, Economics and CS (*with distinction)*

Dissertation Title: *Essays in Applied Microeconomic Theory*

Committee:
Professor Dirk Bergemann (Chair)
Professor Marina Halac
Professor Navin Kartik

Education:
Ph.D., Economics, Yale University, 2027 (expected)
M.Phil., Economics, Yale University, 2024
M.A., Economics, Yale University, 2023
A.B., Mathematics, *magna cum laude*, Princeton University, 2021

Fellowships, Honors and Awards:
Dissertation Fellowship, Yale University, 2026-27
Carl Arvind Anderson Fellowship, Yale University, 2025-26
Benitez Family Fellowship, Yale University, 2023-25
Cowles Foundation Fellowship, Yale University, 2021-26
Phi Beta Kappa, Princeton University, 2021
Sigma Xi, Princeton University, 2021
U.S. Presidential Scholar, 2017

Teaching Experience:
Spring 2026, Teaching Assistant to Prof. Dirk Bergemann, Game Theory (PhD)
Fall 2024, Teaching Assistant to Prof. Ben Polak, Game Theory (Undergraduate / MBA)

October 2026

Research Experience:

Research Assistant to Prof. Dirk Bergemann, Yale University, 2023-25
Research Assistant to Prof. Ryota Iijima, Yale University, 2023
Research Assistant to Sebastian Heise, Federal Reserve Bank of New York, 2020

Publications:

"Screening and Segmenting: A Consumer Surplus Perspective," with Dirk Bergemann and Tibor Heumann, *American Economic Review*, vol. 116, no. 10 (2026): 3768-3801.

"A Unified Approach to Second and Third Degree Price Discrimination," with Dirk Bergemann and Tibor Heumann (2024), *Proceedings of the 25th ACM Conference on Economics and Computation* [extended abstract].

Working Papers:

"A Coasian Theory of Coarse Product Lines," (October 2026), *Job Market Paper*.

"Censorship and Steering in Platform Duopolies," with Kyungho Lee (January 2026).

*"*Optimal Pricing of Cloud Services: Committed Spend Under Demand Uncertainty," with Dirk Bergemann (February 2025).

Seminar and Conference Presentations:

2025-26: AI at Yale Symposium, Microsoft Research New England, Stony Brook ICGT
2024-25: ASSA Annual Meeting, SAET Annual Conference, MURI/TTIC Summer Workshop on Algorithmic Game Theory, Stony Brook ICGT, Asian School in Economic Theory
2023-24: ACM Conference on Economics and Computation

Referee Service:

Quarterly Journal of Economics, AEJ: Micro, AER: Insights, Games and Economic Behavior, ACM Conference on Economics and Computation, ACM-SIAM Symposium on Discrete Algorithms

Military Service:

Captain, New York Army National Guard, United States Army, 2021-present

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References:

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Dissertation Abstract

A Coasian Theory of Coarse Product Lines [Job Market Paper]

Sellers of durable goods—smartphones, cars, software—regularly release improved versions of what they already sell. For such goods, a buyer's purchase decision depends not only on their willingness to pay for quality, but also on the model they currently own, which determines how large an upgrade the new purchase would be. For example, most buyers of the latest iPhone are upgrading from a phone that is at least two years old. These two factors are related: we expect those who buy a new iPhone every year to also value its new features highly. This creates an incentive for sellers to price discriminate on the basis of a buyer's current model. Conditioning prices on behavior, in turn, creates an incentive for buyers to time their purchases strategically. How do sellers design product lines for improving goods when buyers are strategic?

To answer this, I study a monopolist whose production technology improves over time, enabling it to offer progressively higher qualities. Because technology improves without bound, sales recur indefinitely. Buyers are long-lived, and each has a privately known type: their willingness to pay for quality. The seller sets prices over time and may condition them on a buyer's entire purchase history, so in principle the price a buyer faces can depend arbitrarily on their past decisions.

As a benchmark, the socially optimal allocation features complete screening, so higher types upgrade strictly faster and enjoy higher overall quality over time. By contrast, when the seller cannot commit to the price of future goods, it screens coarsely, offering only a few distinct product tiers. Attempts to screen more aggressively are defeated by the buyers' strategic behavior: either buyers wait for a better and cheaper alternative they know is coming soon, or the price is so low that it attracts buyers who would otherwise wait to upgrade. In equilibrium, the seller earns as if all buyers had the lowest valuation, but trade remains inefficient. This contrasts with classic Coasian results, where vanishing rents imply efficiency.

The equilibrium characterization has several implications for product lines of improving goods. Screening is more difficult, and eventually impossible, among higher-type buyers, who upgrade more frequently and so have a stronger incentive to time their purchases. Hence, for any type distribution, the product line features more variety among budget goods. For the same reason, faster technological progress makes screening harder, and eventually impossible, even though more qualities become available. Finally, because screening is coarse, the equilibrium can be implemented with trade-in pricing, in which buyers receive a discount based on their most recent product only. This simple pricing scheme is widely observed in practice, such as in markets for smartphones and cars.

Screening and Segmenting: A Consumer Surplus Perspective, with Dirk Bergemann and Tibor Heumann

We analyze consumer surplus when a monopolist can adjust both prices and product qualities across segments, engaging in second- and third-degree price discrimination simultaneously. We

reduce the problem of finding the consumer-optimal segmentation—an optimization over an infinite-dimensional space—to a tractable optimization problem over a single function subject to a majorization constraint. The solution has a striking structure: consumers with the same value receive the same quality in every segment, though prices differ. Under mild conditions, any segmentation harms consumers if and only if demand is sufficiently more elastic than supply. Hence, potential benefits for consumers depend critically on demand and supply elasticities. These findings have implications for regulatory policy regarding price discrimination and market segmentation.

Censorship and Steering in Platform Duopolies, with Kyungho Lee

We analyze the strategic design of recommendation algorithms in online marketplaces, motivated by stylized facts from Amazon's BuyBox. We show evidence that Amazon strategically suppresses the BuyBox recommendation ("censorship") or selects prices above the minimum available offer ("steering") in a way that closely tracks its largest competitor, Walmart.com. To rationalize these facts, we construct a model of platform duopoly in which two platforms compete by managing buyer search incentives through information design. We characterize the equilibrium and demonstrate a dichotomy: the platform optimally engages in either censorship or steering, but never both simultaneously. In the censorship regime, the platform strategically withholds recommendations to discipline buyer search, effectively raising buyers' expected cost of searching the rival platform. In the steering regime, the platform recommends prices strictly above the competitive level, bounded only by the rival's price, which acts as a stochastic reserve. We show that these algorithmic strategies endogenously replicate the effects of Most-Favored-Nation (MFN) clauses, generating upward pricing pressure on third-party sellers even in the absence of vertical restraints.